Re-measuring the PR #1862 harvest default on m = 3

2026-10-24 · `research/f4_harvest_remeasure_20261024/` · counts only (x86-64, isolation L0) · five sessions of 144 verified records, every audit `OK` with 144 of 144 replays identical

Question and method

PR #1862 made the root linear-row harvest of `koblitz_groebner` the default for inherited F4. The pricing-jump bisect saw about +7% counted solver work on one workload (K0, n = 13). Here the harvest is a method parameter (`linear_harvest = off | default | gpv`, `pdp3-koblitz`), so it can be switched per arm inside ecbench. Five specs (one per curve, 24 public targets each), arms `rho-strong` (reference), `f4-m3-off` (baseline), `f4-m3-default`, `f4-m3-gpv`, and the m = 2 pair as a negative control. Protocol frozen and committed before any session existed.

Results

Control passed: at m = 2, `default / off` = 1.0000 [1.000, 1.000] on all five curves.

curve (dim)	ratio	95% interval	targets up / down	per-target range	mean S off → default	S / rho off → default
n=11, K1 (4)	1.0958	[1.0940, 1.0980]	24 / 0	1.089–1.113	403.5 → 442.1	7.17 → 7.86
n=13, K0 (5)	1.0699	[1.0646, 1.0759]	24 / 0	1.050–1.169	517.0 → 553.1	36.9 → 39.5
n=17, K1 (6)	0.9969	[0.9955, 0.9985]	6 / 18	0.992–1.006	1678.1 → 1672.9	424.1 → 422.8
n=19, K1 (7)	0.9593	[0.9576, 0.9611]	0 / 24	0.953–0.972	3004.4 → 2882.1	1248 → 1198
n=19, K0 (7), XORs unpriced	S 1.0000; band 0.9755	none (point)	S flat	–	4.859 both (lower bound)	1.62 both

Totals ratio of S over 24 paired targets; interval is the harness's cluster bootstrap. `gpv` equals `default` at n = 11, 13, 19 and equals `off` at n = 17, so the n = 17 gain is PR #1891's root-rank gate alone (it fires on none of the other four curves).

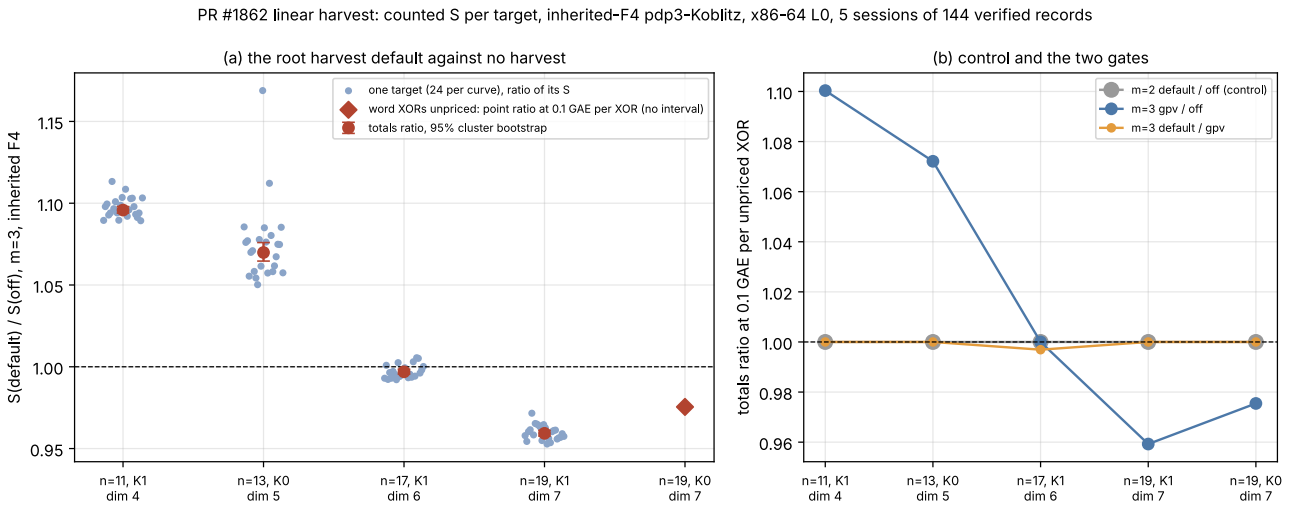


Figure 1: (a) Per-target ratio of S, `default / off` at m = 3 (dots), the harness's totals ratio with its 95% bootstrap interval (red), and the unpriced curve's point ratio at 0.1 GAE per word XOR. (b) The m = 2 control and the two gates. Data: `analysis/*.json` (`ecbench compare --json`); source `plot.py`.

Reading

The sign depends on size: a 7–10% cost at n = 11 and 13 on every one of 24 targets; a 4.1% gain at n = 19 (K1) on every target; a 0.3% gain at n = 17 with mixed per-target signs. Against the current default, switching the harvest off at n = 11 and 13 would lower counted S by about 8.7% and 6.5%. Every priced F4 arm sits 7 to 1,250 times the strong rho in S, and the harvest moves that gap by at most 10%: nothing here brings inherited F4 closer to rho.

Classification and limits

AGENTS.md §3: regression at n = 11 and 13 (counted S and ratio to the floor rise); engineering gain, 4.1%, at n = 19 K1. Not a row of tables A–C, not a speed claim, no wall time; the leaderboard and `benchmarks/` are not updated. Four priced sizes and one unpriced; intervals resample targets within a curve; no exponent fitted. The hypothesis (sign varies with size) was formed after a two-target scratch smoke run; the sessions use different target seeds.

Process facts

The director assessed `executed_with_deviations`: card 1 stopped at a missing `audits/` directory; card 2 was refused (stale pinned commit); card 3 was killed by the executor's 30-minute background limit and left no receipt; cards 4–6 ran one command each, and their executors returned `refused` on a misread stop rule although every command exited 0 and every audit reads OK. All five sessions and audits are intact and none was edited or repeated. Details in the README.